

Titanic Dataset: Visualization Ethics and Communication

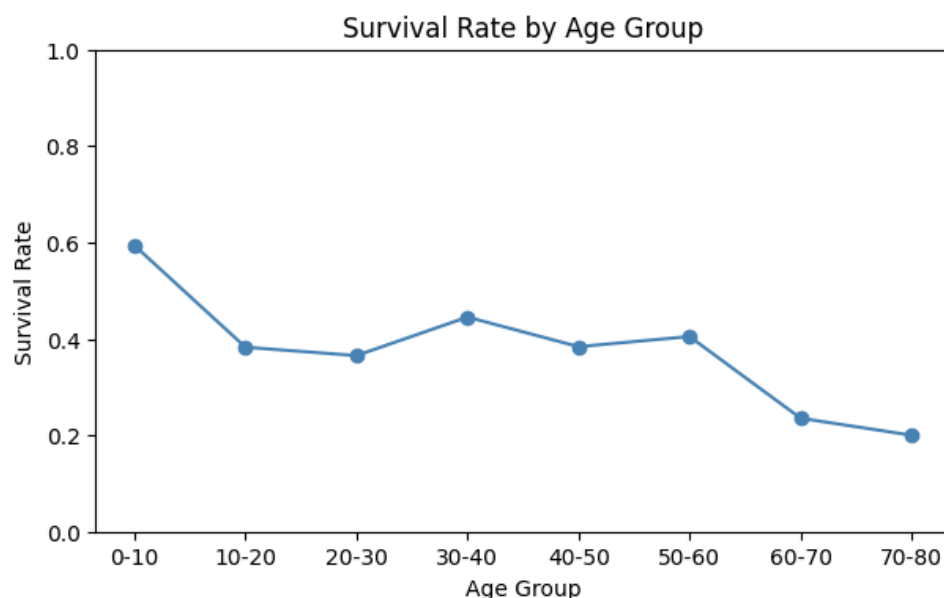
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Introduction: Data Visualization is one of the most important and powerful tools for communicating patterns in data, but it can also be used to mislead the audience, whether intentionally or not. The viewer's conclusion and POV can be changed by the way a chart is designed, which includes the type of chart chosen, how an axis is scaled, the color scheme of the chart, etc. This project aims to explore the Titanic dataset, which has demographic and survival information of 891 passengers including passenger class, sex, age, fare paid and survival outcome.

In this report, we have some accurate visualisations that clearly and honestly communicate the survival patterns in the data followed by some misleading visualisations from the same dataset that are aimed at distorting the views of the audience.

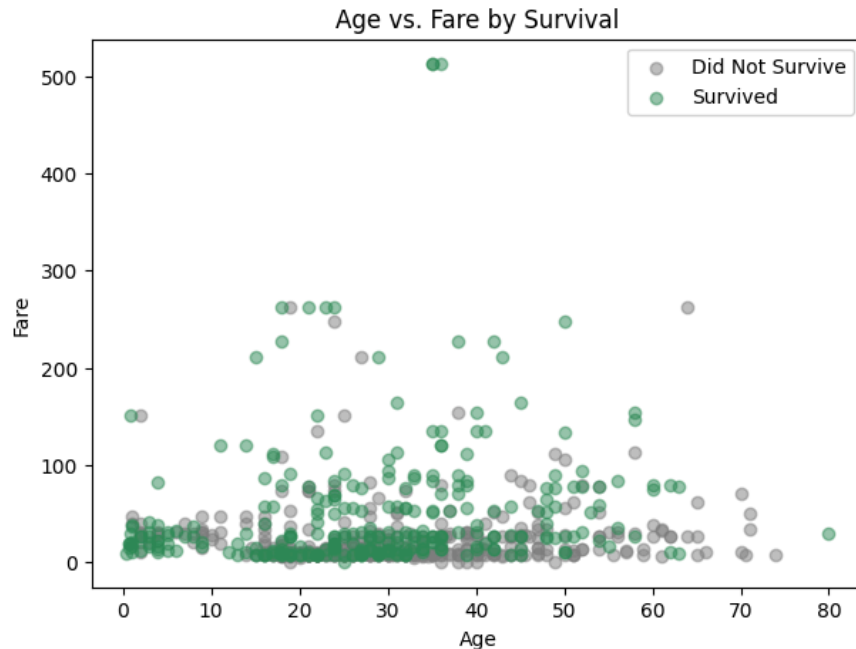
Accurate Visualizations:

- 1. Survival Rate by Age Group:** This visualisation helps us examine how age affected the chances of survival. All the passengers are grouped into 10-year age bands and the average survival rate by each band is calculated, then these results are plotted as a line graph. Line chart was chosen as age bands are ordered and connecting the points allows the reader to follow the trend across all ages rather than comparing individual isolated values. The y-axis ranges from 0%-100% survival rate, so the height and slope of the line accurately reflect the differences between age groups. The chart shows that all the passengers below the age of 10 had a clear survival advantage over other age groups, which is consistent with the evacuation priority for children while the survival rate decreases for passengers over 60.

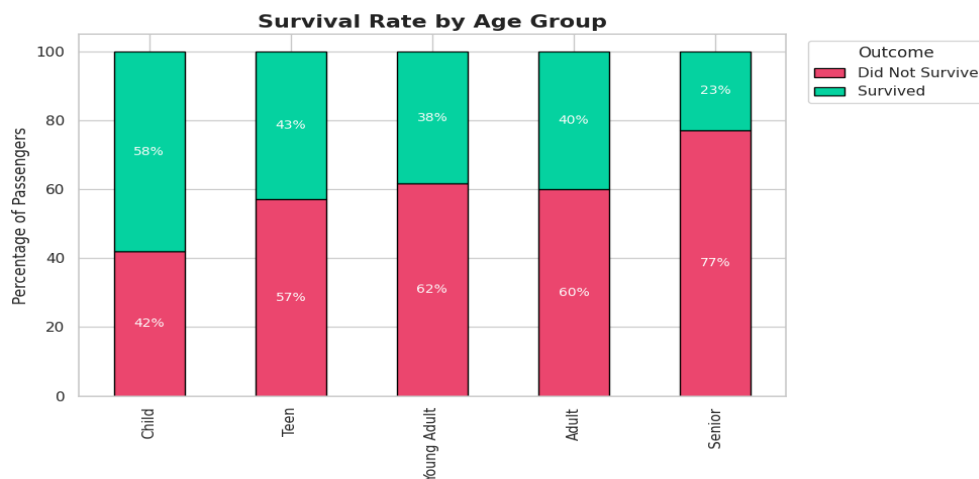


- 2. Age vs. Fare by Survival:** In this visualisation each individual passenger is plotted as a point by age (x-axis) and fare paid (y-axis). Each of these points is colored by whether they survived or not. A scatterplot of the raw data does not hide anything,

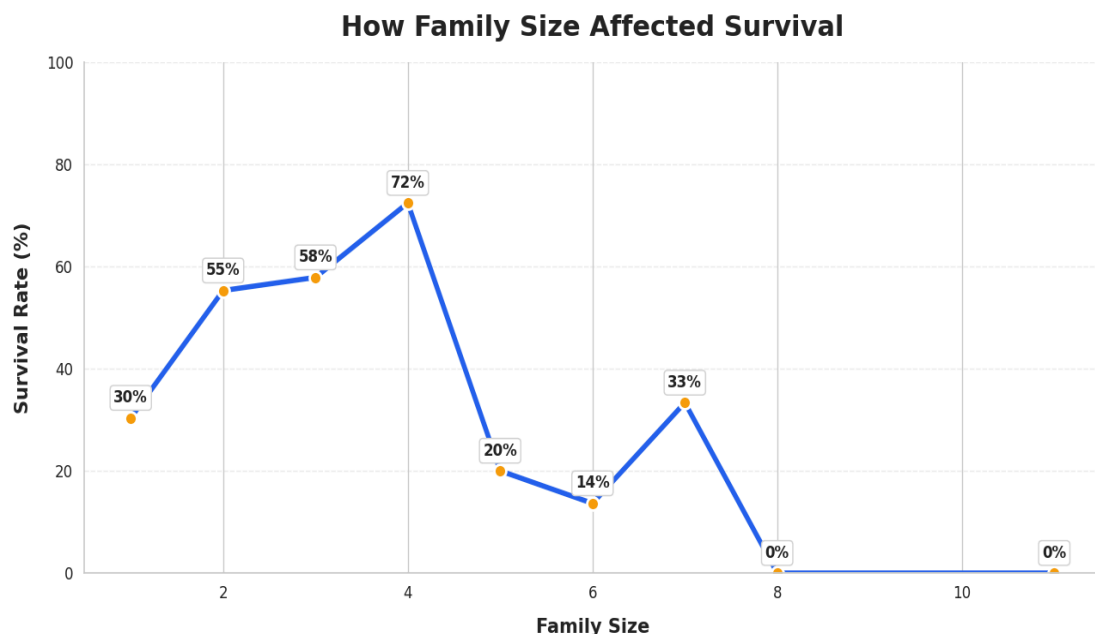
every passenger is represented individually, including outliers and edge cases. All the points are semi transparent which means that overlapping passengers are not obscured and hidden. The chart shows that passengers who paid higher fares had a higher chance of surviving. This chart gives the viewers an honest sense of both pattern and variability.



3. **Survival by Gender:** A survival rate comparison by age groups is represented visually through a 100% stacked bar chart. The illustration enables the user to easily see the proportion of those who survived in each group compared to the ones who didn't. As all bars together equate to the same whole, it becomes obvious that the differences in survival percentages can be compared in a fair manner, not based on the size of those groups. In the chart, children had the highest survival rate with 58%, whereas seniors had the lowest at 23%. Teenagers, young adults, and adults had survival rates of 38% to 43%, respectively. This way, younger passengers were the most vulnerable, i.e. had lesser survival rates. These variations are clearly shown by the stacked format, making the survival outcomes among age groups a much more effective illustration.

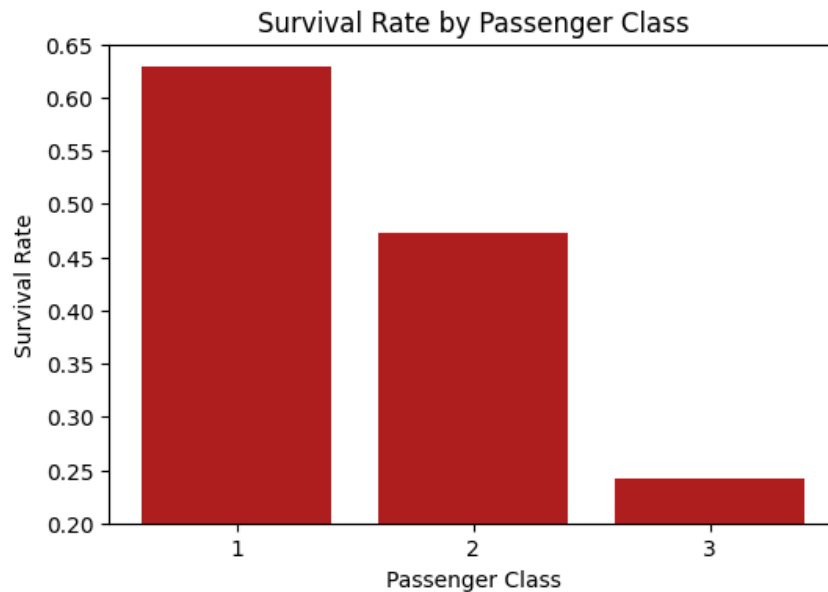


- 4. Family Size and Survival:** This visualization looks at the link family size and passenger survival through a line graph. As family size is an ordered numerical variable, a line graph efficiently illustrates trends in surviving passengers as the number of accompanying family members increases. Research finds passengers with four family members traveling to have the highest survival rate (72%) followed by families with three members (58%) and two (55%) respectively. By contrast, a passenger alone only had a 30% chance of survival whereas very large families were the ones experiencing very low survival rates, with survival dropping to 0% for family sizes of eight and eleven. The clear trend shown by the line makes it easy to notice that passengers traveling in families of moderate-sized generally had the best chance of survival, making this a highly effective visualization for showing the family size - survival relationship.

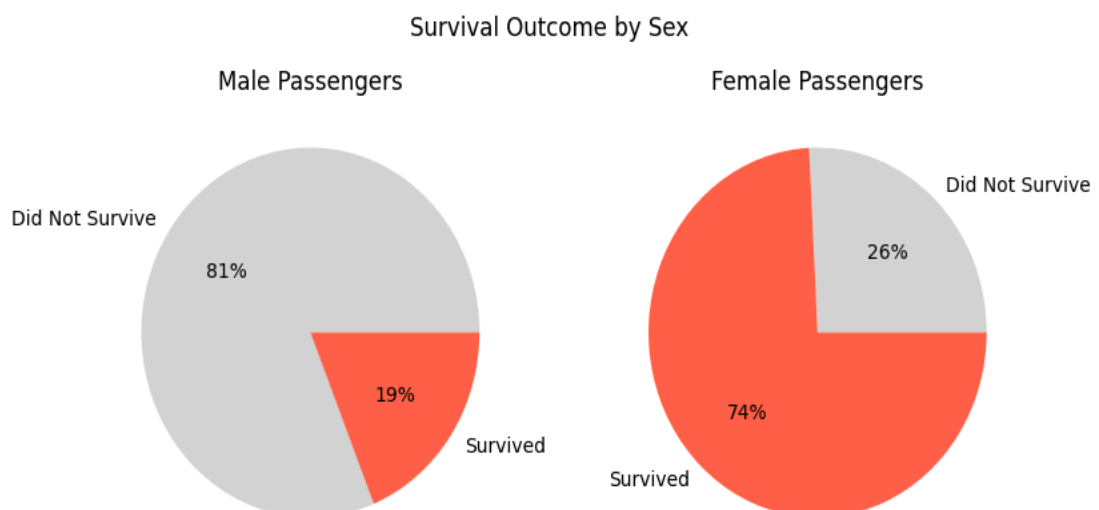


Misleading Visualizations:

- 1. Truncated Y-axis Bar Chart:** This graph uses the exact same accurate survival-rate-by-class dataset as another previous chart, except that the y-axis is truncated from 0 up to 0.20. Since bar charts rely on relative bar height as their primary visual, truncating the y-axis makes the shorter bars shrink more compared to the taller ones. In the data, the 1st class survival rate is 63% which is 2.5 times the 3rd class survival rate (24%), but in the graph the 1st class bar appears 10 times taller than the 3rd class bar. This may mislead the viewers and change their conclusion.

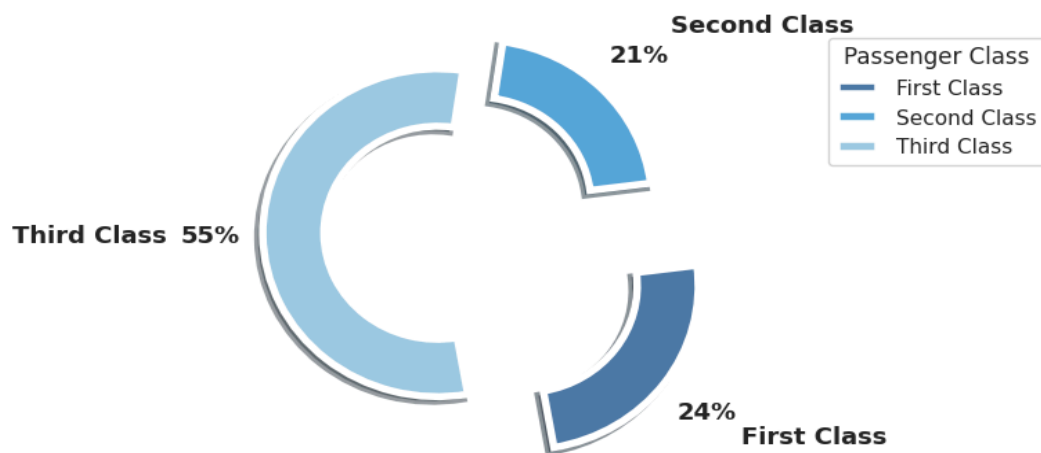


2. **Same Size Pie Charts with Different Totals:** In this visualisation there are 2 pie charts, one for male passengers and one for female passengers, each showing the split between survivors and non survivors. Both the piecharts are of the same size which mislead the viewers to interpret that two same sized pie charts as representing two comparable groups. In the dataset, there are 577 male passengers and only 314 female passengers, nearly twice as many men as women. As a result the female pie charts 74% survival rate appears visually equal in weight to the male's 19% survival rate, even though it represents a much smaller number of people.



3. **Donut Chart of Passenger Class:** The pie chart below illustrates the proportion of passengers in each ticket class as a donut chart. The exact proportions are correct, but due to the shape of the chart it is almost impossible to tell how big one slice is about another, e.g. between first class passengers 24% and second class 21%. The big hole in the center of the chart does not give a better understanding of the data but at the same time takes away a huge part of the screen which was originally allocated for the presentation of the data. the reason that people are much better in judging differences in length rather than angles or areas of the charts. The pie chart might be more attractive visually but in fact it hides rather than reveals the differences between the passenger classes.

Distribution of Titanic Passengers by Class



4. **Sankey Diagram of Gender and Survival:** This Sankey diagram shows the movement of passengers from gender to survival outcome. It highlights correctly that most male passengers perished, whereas a bigger proportion of female passengers were saved. Still, overlapping flows and change in widths hinder clear comparison of the values between the groups. Even though the appearance is quite stunning, the diagram makes the simple comparison unnecessarily complex. The same comparison can be made very conveniently by a double bar chart. The viewers can easily read off the survival outcomes by gender there without having to track several flow paths. So, presenting the dataset through the Sankey diagram would not be the case even though it was based on the correct data.

Titanic Survival Flow by Gender



Discussion: The four effective visualizations conveyed key patterns quite clearly using matching chart types. For example, the 100% stacked bar chart quickly demonstrated the survival rates in different age groups. The line chart illustrated how survival varied with family size quite nicely. The scatter plot showed age vs fare, without concealing age-fare outliers. Finally, the gender comparison chart highlighted the survival disparity between men and women very clearly. Via these means, viewers became able to recognize significant patterns almost immediately. The misleading visualizations, then again, revealed that it is quite wrong not to design properly when you want to present data even when it may not be so bad after all. The truncated y-axis bar chart overstated the difference in the classes of passengers. The pie charts having equal size forgot that the passenger groups were not at the same size. The donut charts hindered more than helped the comparison of classes of passengers. The Sankey diagram was overcomplicated for just a comparison of things. Such examples only confirm one more time that visualization design can change a viewer's perception of the truth without even altering the data.

Conclusion: This project shows that good data visualization is not only the representation of nice-looking pictures, but accurate information communication. It's been demonstrated that the better-designed visualizations not only drew attention to the main points, like a child's survival rate which was 58%, passengers traveling with four family members had 72% survival rate, and survival rates being higher among female first-class passengers, but also did so clearly and straightforwardly. These discoveries would have been harder to find if the graphs had not been picked as the nature of the dataset. The other charts which misled readers included only data, but they did confuse rather than clarify. According to whether the viewers were presented with skewed graphs, improper types of chart, or superfluous visual complexity, each case illustrated the ways in which viewers may be led to different interpretations based solely on the way information is given. The lesson is that when it comes to choosing visualizations we should focus on the main goals of the visualizations which are

to be clear, to communicate truthfully, and to be accurate. Generally speaking, this project stresses that doing data visualization in an ethical way involves not only putting out correct data, but also presenting it in such a way that the viewers can interpret the results without bias and with the knowledge necessary.